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Two-component, *ab initio* potential energy surface for CO₂—H₂O, extension to the hydrate clathrate, CO₂@(H₂O)₂₀, and VSCF/VCI vibrational analyses of both

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We report an *ab initio*, full-dimensional, potential energy surface (PES) for CO₂—H₂O, in which two-body interaction energies are fit using a basis of permutationally invariant polynomials and combined with accurate potentials for the non-interacting monomers. This approach which we have termed “plug and play” is extended here to improve the precision of the 2-body fit in the long range. This is done by combining two separate fits. One is a fit to 47 593 2-body energies in the region of strong interaction and approaching the long range, and the second one is a fit to 6244 2-body energies in the long range. The two fits have a region of overlap which permits a smooth switch from one to the other. All energies are obtained at the CCSD(T)-F12b/aug-cc-pVTZ level of theory. Properties of the full PES, i.e., stationary points, harmonic frequencies of the global minimum, etc., are shown to be in excellent agreement with direct CCSD(T)-F12b/aug-cc-pVTZ results. Diffusion Monte Carlo calculations of the dimer zero-point energy (ZPE) are performed, and a dissociation energy, D_0 , of 787 cm⁻¹ is obtained using that ZPE, D_e , and the rigorous ZPEs of the monomers. Using a benchmark D_e , D_0 is 758 cm⁻¹. Vibrational self-consistent field (VSCF)/virtual state configuration interaction (VCI) MULTIMODE calculations of intramolecular fundamentals are reported and are in good agreement with available experimental results. Finally, the full dimer PES is combined with an existing *ab initio* water potential to develop a potential for the CO₂ hydrate clathrate CO₂(H₂O)₂₀(5¹² water cage). A full normal-mode analysis of this hydrate clathrate is reported as are local-monomer VSCF/VCI calculations of the fundamentals of CO₂. Published by AIP Publishing. [<http://dx.doi.org/10.1063/1.4994543>]

I. INTRODUCTION

The Force Field (FF), the focus of this special issue of J. Chem. Phys., is a generic term that denotes an analytical representation of a potential describing the interactions of a molecule or a molecular cluster. The molecule may be as small as a triatomic or as large as a huge covalently bound polymer. A molecular cluster is, as the name implies, a collection of molecules bound by weaker, so-called non-covalent interactions. Typically, FFs for molecular clusters make use of the rigid-monomer approximation. This leads to a great simplification in the complexity/dimensionality of the FF. In the general case of flexible monomers, the FF is typically decomposed as a sum of intramolecular and intermolecular terms. The intramolecular terms are typically expressed as bond stretch and bending functions (often harmonic) and dihedral (torsional) interactions. The intermolecular interactions are typically expressed as Lennard-Jones or “exp-6” potentials, augmented by two-body multipole interactions, see, for example, “CHARMM.”¹ More sophisticated FFs do account for many-body induced interactions as well, for example, the “AMOEBA” water potential.² These interactions contain linear and non-linear parameters that are typically determined empirically, e.g., using known equilibrium structures,

vibrational frequencies, thermodynamic properties, etc., or by using *ab initio* electronic energies to obtain the parameters by some fitting procedures, typically a linear/non-linear least square optimization. Generally this approach does not provide precise fitting. Obtaining RMS fitting errors with tens to hundreds of cm⁻¹ is a typical result for FFs with flexible monomers. As a result, structures and normal mode frequencies typically differ by few percent relative to the *ab initio* results. Of course, by introducing more linear and non-linear parameters, the fitting can be improved. Nevertheless, this approach has several limitations. One is the imposition of the models for intra- and intermolecular interactions. This is especially true for intramolecular interaction, where harmonic models are often used. It is also true for the short-range parts of the intermolecular, which are typically represented by exponentials. Second, these FF representations generally do not describe three and higher-body interactions, and they do not describe chemical reactions or isomerization (there are some exceptions, such as Reaxff³).

In the past decade or so, essentially model-free approaches to the analytical representation of potential energy surfaces have been developed and applied to numerous molecular systems including chemical reactions. These PESs are numerical fits to tens of thousands of high-level electronic energies. Ours in particular make use of fitting bases that are invariant with respect to permutations of like atoms. These general

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approaches have been reviewed,^{4–7} and the interested reader is directed to these.

A particular subset of these approaches, suitable for large molecular clusters governed by non-covalent interactions, has been developed by our group building on the seminal work of others,⁸ albeit for rigid monomers. In the general approach for flexible monomers, the intermolecular interaction is represented by a many-body expansion with the important 2- and 3-body fits obtained by the numerical approaches mentioned above. For example, for flexible water, these representations, which are high-dimensional, have been determined based on fitting tens of thousands of high-level electronic energies.^{9,10} These representations of the water potential are transferable to other systems such as hydrate clathrates, as we have recently reported for the hydrate clathrates of H₂¹¹ and CH₄.^{12,13} In these examples, 2- and 3-body, full-dimensional potentials were developed for H₂—H₂O and CH₄—H₂O and combined with the water potential and pre-existing high-level monomer potentials to obtain the full PES. We have termed this approach wherein existing monomer potentials are combined with fitted 2-body (and 3-body) potentials “plug and play.”

All of these numerical-based approaches produce highly precise fits, i.e., of the order of 10–100 cm^{−1} RMS fitting error, depending on the total energy span of the dataset. However, even a 10 cm^{−1} RMS fitting error would be considered large in the long-range part of the potential, where the interaction energies are of the order of tens of wavenumbers or less. In this paper, we report a new 2-body potential for CO₂—H₂O and combine it with high-level monomer potentials to obtain the full dimer PES. The approach described in those papers is extended here to greatly improve the precision of the 2-body PES in the long range, in recognition of the comments made above.

The CO₂—H₂O dimer has been extensively studied both experimentally and theoretically, as it is important for understanding the basis of many fundamental reactions such as photosynthetic conversion from CO₂ and H₂O to organic molecules. The complex has been proposed to be present in some planetary atmospheres.¹⁴ In addition, the CO₂ hydrate clathrate, whose structure will be described later, has been proposed in the Martian atmosphere^{15–18} and also as a possible means to sequester CO₂.^{19,20}

Early experiments on the CO₂—H₂O complex determined the minimum energy configuration. This was first reported by Peterson and Klemperer in a molecular-beam-electric resonance experiment.²¹ With the determination of rotational constants, hyperfine spectra, and dipole moments, it was shown that the global minimum geometry is of C_{2v} symmetry [Fig. 4(a)]. Later, this result was confirmed and supported by further experiments.^{22–24}

Many *ab initio* studies of stationary points and energetics of this complex have been reported over the past 40 years.^{25–34} The complex is a challenge for theory because of the weak binding and the floppy nature of the stationary configurations. After several thorough studies using high-level methods, stationary points have now been well-characterized,^{33,35} and the early reports of two minima are not correct.

Developing an *ab initio*-based, full-dimensional, potential energy surface for the CO₂—H₂O complex is clearly desirable, albeit a challenge owing to the high dimensionality (12 vibrational degrees of freedom). Several PESs with rigid monomers, which reduce the dimensionality to 5, have been reported.^{35,36} The most recent rigid-monomer PES was reported by Makarewicz in 2010.³³ This PES was a fit to roughly 23 000 using MP2/aug-cc-pVTZ (plus bond-centered functions) energies, using a complex analytical expression that included damped long-range electrostatic and dispersion interactions. 168 non-linear parameters were optimized to obtain a precise fit. The reported unweighted RMS fitting error for 22 500 configurations up to an energy of 3500 cm^{−1} is 11.8 cm^{−1}. This PES is certainly of high quality and can be used in a number of studies, but the limitation to rigid monomers certainly rules out applications to IR spectroscopy.

To the best of our knowledge, the only previous *ab initio*-based PES for flexible monomers was reported in 2005 by Wilcox and Bauer.³¹ They used the conventional FF to represent this potential, and it is worthwhile to recall this approach in detail. For the two intramolecular terms, they used following equations to represent the CO₂ potential and H₂O potential:

$$E_{\text{CO}_2} = \frac{1}{2}k_1(r_{\text{C-O}_1} - 1.169\,14)^2 + \frac{1}{2}k_1(r_{\text{C-O}_2} - 1.169\,14)^2 + \frac{1}{2}k_{12}(r_{\text{O}_1\text{-O}_2} - 2.338\,27)^2 + \frac{1}{2}k\delta(a_{\text{OCO}} - \pi)^2, \quad (1)$$

$$E_{\text{H}_2\text{O}} = \frac{1}{2}k_1(r_{\text{O-H}_1} - 0.968\,76)^2 + \frac{1}{2}k_1(r_{\text{O-H}_2} - 0.968\,76)^2 + \frac{1}{2}k_{12}(r_{\text{H}_1\text{-H}_2} - 1.523\,26)^2 + \frac{1}{2}k\delta(a_{\text{HOH}} - 1.809\,17)^2, \quad (2)$$

where k_1 , k_2 , k_{12} , and $k\delta$ are parameters, r_{a-b} are 6 intramolecular atom(a)-atom(b) bond lengths (three for each monomer) in Å, and a_{OCO} and a_{HOH} are the bond angles in radians. They expressed the intermolecular interactions with the form,

$$E_{\text{intermolecular}} = \sum_{i=1}^{n\text{-set}} \sum_{j=1}^9 C(i, j)(1/r_j)^i, \quad (3)$$

where for simplicity j represents 9 intermolecular atom-atom pairs. Here “ n -set” determines the number of parameters. The best model has a RMS fitting error of 0.34 kcal/mol and contains two minima, which are (incorrectly) present in the dataset of 172 G3(MP2)//B3LYP energies. The potential at the T-shaped global minimum is −2.84 and −3.18 kcal/mol for G3(MP2)//B3LYP and the force field, respectively. Another important limitation of the FF is that it is valid only for separations of the water oxygen and the carbon atom over the range

of 2.0–8.0 Å. Thus, it cannot describe long-range interactions. Finally, given the limited range of energies in the dataset (up to 10 kcal/mol relative to separated monomers at equilibrium), it is unlikely that this force field could yield accurate anharmonic vibrational energies. In this context, the work of Chaban *et al.* is noteworthy.¹⁴ This group reported anharmonic calculations of the monomer vibrational energies by obtaining electronic energies directly on grids and thus without recourse to an existing PES. We discuss these calculations in detail below.

The PES we report is based on fits to tens of thousands CCSD(T)-F12b/aug-cc-pVTZ 2-body interactions with flexible monomers. We use a fitting basis of permutationally invariant polynomials in the so-called Morse variables. Details of this approach have been given elsewhere,^{4,37} and so only a brief review of the essential aspects is given below. An important feature of the current approach is the use of two fits to create the final 2-body PES. One is a fit in the strong interaction region and in the moderate long-range region and the other is a fit for a much longer-range region. We denote these fits as “fit-SR” and “fit-LR.” These two fits are then combined using a simple switching function to obtain the final 2-body PES. The 2-body PES is combined with the previous 1-body monomer potential energy surface to obtain the full potential energy surface for the dimer. In addition, these monomer 1-body and 2-body potentials are combined with an accurate *ab initio* potential for water to develop a potential for CO₂ hydrate clathrates. These PESs are then used in a variety of anharmonic vibrational calculations.

The rest of the paper is organized as follows. In Sec. II, computational details of the *ab initio* calculations, fitting methods, diffusion Monte Carlo, vibrational self-consistent field (VSCF)/virtual state configuration interaction (VCI) calculations, and the many-body representation of the CO₂ hydrate clathrate potential are given. Results and discussion are given in Sec. III. These include an analysis of the PES, harmonic and VSCF/VCI vibrational energies for the dimer, followed by analyses of the hydrate clathrate calculation including a full normal-mode analysis and local monomer calculations of the fundamentals of intramolecular modes of CO₂. Finally, a summary and conclusions are given in Sec. IV.

II. THEORY AND COMPUTATIONAL DETAILS

The “plug and play” approach^{11–13} for CO₂–H₂O consists in writing the full potential as

$$V_{\text{CO}_2\text{--H}_2\text{O}} = V_{\text{CO}_2}^{(1)} + V_{\text{H}_2\text{O}}^{(1)} + V_{\text{CO}_2\text{--H}_2\text{O}}^{(2)}, \quad (4)$$

where $V_{\text{CO}_2}^{(1)}$ and $V_{\text{H}_2\text{O}}^{(1)}$ are the monomer potentials and $V_{\text{CO}_2\text{--H}_2\text{O}}^{(2)}$ is the intrinsic two-body potential. Hereafter, we refer this full potential as the “PES.” The appeal of this approach is that the monomer potentials are pre-existing ones, which can be chosen with a particular application in mind. For the present purpose, which is focused on vibrational dynamics, we selected monomer potentials that are of spectroscopic accuracy, when used in essentially exact vibrational calculations. Thus, the one for H₂O is from the work of Partridge and

Schwenke³⁸ and the one for CO₂ is from the work of Huang *et al.*³⁹

The objective is to obtain a full-dimensional potential energy surface for $V_{\text{CO}_2\text{--H}_2\text{O}}^{(2)}$; we will refer to this as the “PES_{2b}.” This is done by fitting a database of *ab initio* interaction energies obtained from three electronic structure calculations at a given nuclear configuration, one for the complex ($V_{\text{CO}_2\text{--H}_2\text{O}}$) and one each for the non-interacting monomers ($V_{\text{CO}_2}^{(1)}$ and $V_{\text{H}_2\text{O}}^{(1)}$). Because this interaction energy goes to zero as the monomers separate, the final PES_{2b} should faithfully reproduce this and also be as relatively precise in the long range as in the short range. In order to meet these requirements, PES_{2b} is a hybrid of two fits, which we denote as fit-SR and fit-LR, and which are joined by a simple switching function. The first fit, as the name implies, is focused on the short to moderate long range, while fit-LR is focused on the long range and rigorously goes to zero as monomers separate.

A. *Ab initio* electronic energies databases

In order to sample the high-dimensional configuration space of interest, we first performed “*ab initio* molecular dynamics” calculations at the MP2/VDZ level of theory, with initial starting configurations at known stationary points and with initial kinetic energies ranging from 50 to 3000 cm^{−1}. From these calculations, roughly 21 000 widely spread configurations were saved for higher level calculations of energies. Additional configurations were obtained based on the database of 23 050 configurations used by Makarewicz³³ for rigid monomers. Those configurations were generated using grids in the CO(water) distance (from 2 to 10 Å) and four angles specifying the orientation of the two rigid monomers. We used those configurations but with a small, random, displacement of a monomer bond length (±0.005 Å) or bond angle (±3°) at each. Finally, for the first generation dataset, 1000 additional configurations were added along an approximate least-motion dissociation path and also a similar internal rotation path starting at the global minimum. In this way, the first-generation database of roughly 45 000 configurations was established.

Electronic energies at these configurations were obtained using the CCSD(T)-F12b theory,^{40,41} as implemented in MOLPRO,⁴² with an aug-cc-pVTZ basis.^{43,44} Based in part on a previous study of CO₂–H₂O, where it was shown that the electronic binding energy using CCSD(T)-F12/VDZ was within 0.05 kcal/mol of the converged result (with respect to the basis size),³⁴ no corrections to possible basis-set superposition errors were done.

After obtaining an initial fit (see below for details) using this database, unbiased diffusion Monte Carlo^{45–47} calculations for the zero-point state of the complex were performed in order to explore regions of the fit that might produce unphysical large negative values, i.e., “holes.” A number of such “holes” were found and in their vicinity roughly 2000 electronic energies were added to the database. This database was used to obtain fit-SR (details given below). Finally, for fit-LR, 6244 long-range configurations were sampled with R_{CO} between 6 Å and 22 Å, where R_{CO} is the

distance between carbon and the oxygen atom of H₂O. The database consists of all (1280) configurations and corresponding *ab initio* energies in the large database for fit-SR for R_{CO} is greater than 6 Å, plus 4964 additional configurations for R_{CO} in the range 6–22 Å for various internal configurations of the two monomers and relative orientations of the monomers.

B. PES_{2b} fitting

The approach to obtain fit-SR and fit-LR is the usual one we take, that is using a basis of permutationally invariant polynomials in Morse variables.⁴ CO₂–H₂O is of the form $X_3Y_2Z_1$ and thus the full symmetric group is of order $3!2! = 12$. The Morse variables are $y_{ij} = \exp(-r_{ij}/\lambda)$, where r_{ij} is the internuclear distance between atom i and atom j and λ is a range parameter. For fit-SR, λ is the (usual) value of 2.0 a.u. For fit-LR, after some optimization, a larger value, $\lambda = 7.0$ a.u., is used.

For fit-SR, the polynomial basis consists of a group of factored primary and secondary invariant polynomials,⁴ namely,

$$V(\mathbf{y}) = \sum_{\alpha=0}^M h_{\alpha}(\mathbf{p}(\mathbf{y}))q_{\beta}(\mathbf{y}), \quad (5)$$

where α is the total polynomial order of each term in the sum, starting from 0 to maximum order of M . Each order of polynomial itself is a summation of a series of symmetrized polynomials determined by polynomials h_a and secondary invariant polynomials q_b . Moreover, h_a is a polynomial of primary invariants $p(\mathbf{y})$. The number of $p(\mathbf{y})$ is equal to the number of Morse variables, which is 15 for this six-atom system. In the present case, the maximum polynomial order is 6 which results in 5835 linear coefficients after expansion. The unknown coefficients were obtained using the standard least-squares method.

As was noted in the review on the above fitting method,⁴ the above factored expression does not rigorously separate into non-interacting fragments in the corresponding asymptotic limit. Instead this asymptotic behavior is obeyed numerically by the fit. The reason for this non-separability and the remedy will become clear below, where we present a second permutationally invariant fitting method for fit-LR. This method is a modified version of monomial symmetrization (MSA).^{4,37} In this approach, the potential is represented in basis of symmetrized monomials, namely,

$$V(\mathbf{y}) = \sum_{m=0}^M D_L \mathcal{S} \left[\prod_{i < j}^N y_{ij}^{l_{ij}} \right], \quad (6)$$

where $N = 15$ is the number of Morse variables and D_L are the linear unknown coefficients. Each monomial is symmetrized by formal operator $\mathcal{S}$ based on the relevant permutation group.^{4,37} Note that for a given maximum polynomial order, the number of unknown coefficients in the two approaches is equal, as the two approaches are formally equivalent. The modification mentioned above is to “purify” the basis so that the expression rigorously goes to zero as the monomers separate.¹¹ This is done by eliminating terms in the basis that contains only intramolecular monomer Morse

variables. For the long-range fit to the database of roughly 6000 electronic energies, the maximum polynomial order is 4, which results in 524 coefficients. The purification procedure eliminates 103 terms resulting in 421 coefficients finally.

Next to combine fit-SR and fit-LR to generate PES_{2b}, we use a simple, finite-ranged switching function in the CO distance.^{9,48–50} Thus,

$$\text{PES}_{2b} = (1 - s(R_{\text{CO}}))(\text{fit-SR}) + s(R_{\text{CO}})(\text{fit-LR}), \quad (7)$$

where s is given by

$$s(R_{\text{CO}}) = \begin{cases} 0, & \text{if } R_{\text{CO}} < a \\ 10p^3 - 15p^4 + 6p^5, (p = \frac{R_{\text{CO}} - a}{b - a}) & \text{if } a \leq R_{\text{CO}} \leq b \\ 1, & \text{if } R_{\text{CO}} > b \end{cases} \quad (8)$$

Thus, s is a smooth function that varies from 0 to 1 in the switching region $[a, b]$. Notice that PES_{2b} will be reduced to fit-SR when $R_{\text{CO}} < a$ and fit-LR when $R_{\text{CO}} > b$. In the present application, the switching region is from 5 to 6 Å. This range was determined empirically by inspecting potential cuts in R_{CO} , as shown below, and locating a region where the cuts from the two fits overlap well.

C. Dissociation energy and vibrational analysis of the CO₂–H₂O dimer

The rigorous zero-point energy (ZPE) was obtained using the unbiased diffusion Monte Carlo (DMC) method.^{45–47} In brief, this method uses a stochastic approach to solve the Schrödinger equation in imaginary time in Cartesian coordinates. The long-time solution decays to the zero-point state, and thus the method provides a way to determine both the wavefunction and energy of this state. In our application, 10 DMC trajectories were run for both the complex and the monomers. For each trajectory, 30 000 walkers were propagated for 50 000 steps with the last 40 000 used to determine the wavefunction and zero-point energy. The number of potential calls per trajectory is of the order of 10^{10} , and clearly, a PES is essential for this approach. Even with a PES, the central processing unit (CPU) effort for these calculations is of the order a couple of days on a single node of a workstation. With these rigorous ZPEs, combined with an accurate electronic dissociation energy, D_e , a rigorous dissociation energy D_0 was determined.

For excited vibrational states, the code MULTIMODE^{51–55} was used to perform vibrational self-consistent field (VSCF) and virtual state configuration interaction (VCI) calculations.^{55,56} There are 12 vibrational modes of this floppy complex, and an exact treatment of all of these is beyond the current state of the art. Thus, we restrict attention to the seven intramolecular modes. Even with this reduction, the potential is 7-dimensional and doing numerical quadratures of matrix elements in this dimensionality would be very compute intensive. To overcome this problem, MULTIMODE uses a hierarchical n -mode representation of the potential.⁵¹ Based on many previous calculations, a 4-mode representation (4MR) of the potential should produce results that are converged to within a few wavenumbers or less. With this 4MR, the maximum quadrature dimensionality is 4; however, there are $\frac{7!}{4!3!} = 35$

4-mode grids. For the basis, a direct-product is used and the single-mode basis consists of an underlying primitive basis of 12 harmonic-oscillator functions. In the VCI calculation, the excitation space consists of single, double, triple, and quadruple excitations with the sum of quanta for these levels of excitations being 12, 11, 10, and 9, respectively. This resulted in a VCI matrix of order 26 496, which is readily diagonalized.

D. CO₂ hydrate clathrate

There is great interest in CO₂ hydrate clathrates with the corresponding extensive literature, and it is certainly beyond the scope of this paper to review this literature. Our purpose here is to illustrate how the new CO₂–H₂O PES can be used to develop the many-body potential for these clathrates. We do note that there have been several studies of the dynamics of CO₂ in water cages making use of empirical interaction potentials.^{57,58} Density functional theory, B3LYP/6-31G(d), was used to obtain the optimum and maximum cage occupancy for five different cages.⁵⁹

The lack of an *ab initio*-based clathrate model motivates us to construct a many-body-representation based high quality PES for CO₂ hydrate clathrates. Thus, we use the well-known many-body representation of the potential given by

$$\begin{aligned}
 V_{\text{CO}_2 @ (\text{H}_2\text{O})_n} \approx & V_{\text{CO}_2}^{(1)} + \sum_{i=1}^n V_{\text{H}_2\text{O}(i)}^{(1)} \\
 & + \sum_{i=1}^n V_{\text{CO}_2-\text{H}_2\text{O}(i)}^{(2)} + \sum_{i=1}^n \sum_{j=i+1}^n V_{\text{H}_2\text{O}(i)-\text{H}_2\text{O}(j)}^{(2)} \\
 & + \sum_{i=1}^n \sum_{j=i+1}^n \sum_{k=j+1}^n V_{\text{H}_2\text{O}(i)-\text{H}_2\text{O}(j)-\text{H}_2\text{O}(k)}^{(3)} \\
 & + V(\text{higher order water potential terms}), \quad (9)
 \end{aligned}$$

where n is the number of water molecules. For the water potential, we use the accurate *ab initio* MB-pol potential,¹⁰ which also uses the Partridge-Schwenke H₂O monomer potential. (We used this many-body representation for the CH₄ hydrate clathrate,^{13,60} with the *ab initio* WHBB water potential.⁶¹)

The above representation is restricted to 2-body CO₂–H₂O(i) interactions, and we use the one we are reporting here. In line with numerous previous studies, we expect the 3-body interaction $V_{\text{CO}_2-\text{H}_2\text{O}(i)-\text{H}_2\text{O}(j)}^{(3)}$ to be much less than the 2-body $V_{\text{CO}_2-\text{H}_2\text{O}(i)}^{(2)}$ one. However, in a hydrate clathrate, there are more 3-body interactions than 2-body ones so it is necessary to examine the total interaction in detail. Thus, we calculated these two potentials at the optimized global minimum geometry of CO₂(H₂O)₂₀ at the MP2/aVTZ level of theory. For this hydrate clathrate, there are 20 $V_{\text{CO}_2-\text{H}_2\text{O}(i)}^{(2)}$ terms and 190 $V_{\text{CO}_2-\text{H}_2\text{O}(i)-\text{H}_2\text{O}(j)}^{(3)}$ terms, and thus the total 3-b and 2-b potentials are -100.7 cm^{-1} and -3783.2 cm^{-1} , respectively. Thus, the total 3-body potential is about 2.7% of the total 2-body potential and so we can safely ignore the 3-body interactions in this study. Finally, we denote this many-body potential as PES-CW20, as “C” stands for “Carbon dioxide” and “W” for “Water.”

Two vibrational analyses of CO₂@(H₂O)₂₀ are done, following a geometry optimization. One is a full normal-mode analysis of the hydrate clathrate and the second one is a VSCF/VCI local monomer (LMon) analysis of the embedded CO₂. In the LMon approach,⁶¹ the motion of flexible CO₂ is considered with the full interaction of the water molecules, which are fixed at their reference configurations at the minimum of the hydrate clathrate. In this approach, a normal-mode analysis is done for the embedded CO₂; we refer to this as “local normal-mode” analysis. This yields 9 non-zero frequency modes, five of which are frustrated rotation and translational modes and 4 are the intramolecular modes. MULTIMODE VSCF/VCI calculations are done using the 9 local modes and also the 4 intramolecular modes. For both, a 4MR of the potential is used. For these calculations, each single mode function is represented by a basis of 9 harmonic-oscillator wavefunctions. In the VCI calculation, the excitation space again consists of single, double, triple, and quadruple excitations, with the sum of quanta for these levels of excitations being 9, 8, 7, and 6, respectively. The VCI matrices are of order 24 310 and 715 for the 9-mode and 4-mode calculations, respectively.

III. RESULTS AND DISCUSSION

A. PES_{2b} fitting precision

The distribution of 2-body energies is shown in Fig. 1. As seen, there is a concentration of energies around 0 cm^{-1} . This comes mostly from the fit-LR database. Also, a majority of energies are negative, and this is mainly due to *ab initio* molecular dynamics (AIMD) sampling, which favors configurations with low potential energy. The cutoff at around -1000 cm^{-1} is a rough indication of the electronic binding energy of the dimer. The positive 2-body energies are indicative of repulsive interactions, which must be included to obtain a global 2-body PES.

The RMS fitting error for fit-SR to the full database of roughly 48 000 energies is 6.3 cm^{-1} and the one for fit-LR is 0.40 cm^{-1} . The RMS for all negative energies is 5.17 cm^{-1} (34 119 configurations) for fit-SR and 0.32 cm^{-1} (4890 configurations) for fit-LR. This clearly indicates precise fitting,

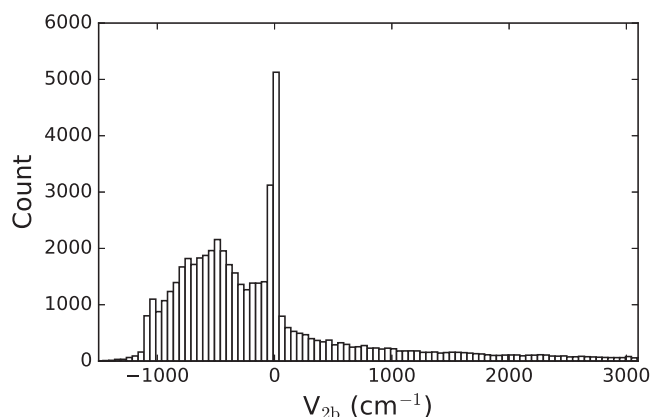


FIG. 1. Histogram distribution of 2-body energies V_{2b} from the combined datasets (roughly 53 000 energies) for fit-SR and fit-LR. Bin width is 50 cm^{-1} .

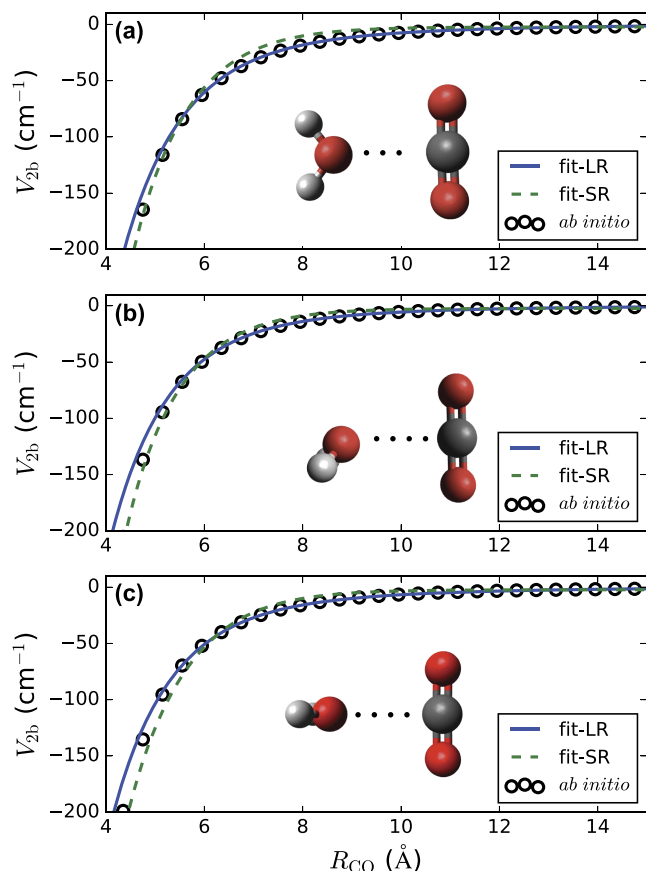


FIG. 2. Rigid monomer dissociation cuts starting from the equilibrium structure along the R_{CO} direction. Panels (a)–(c) correspond to geometries in Figs. 4(a)–4(c).

especially for the long-range fit. Thus, the objective of obtaining a small RMS fitting error for the long range has been achieved. In addition, fit-LR goes to zero rigorously as the monomers separate.

We next examine the two fits and direct *ab initio* calculations for three cuts as a function of R_{CO} in Fig. 2. As seen, and as expected, fit-SR matches the *ab initio* energy very well up to about 6 Å, while fit-LR is almost identical to *ab initio* energies from 6 Å. Also, and of direct relevance to the switch between these two fits, we observe that the two fits are in very

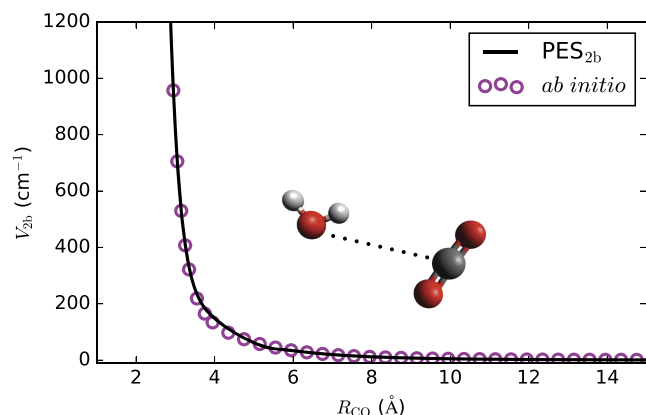


FIG. 3. Rigid monomer potential cut versus R_{CO} for the indicated repulsive configuration.

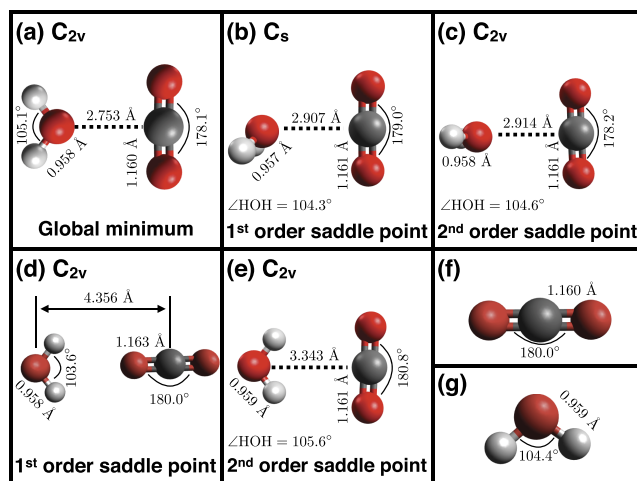


FIG. 4. (a)–(e) are geometries of saddle points obtained from PES. Selected bond lengths and angles are shown directly in the figure. (f) and (g) are monomer global minimum configurations obtained from monomer PESs. Note that our geometries (d) and (e) correspond to geometries (f) and (d), respectively, in Ref. 33.

good agreement in the range 5–6 Å. Thus, the switching region is 5–6 Å.

The accuracy of the repulsive interaction between CO_2 and H_2O in the fitted PES_{2b} is also important to examine. This is done in Fig. 3 for the indicated monomer orientation, where good agreement with direct *ab initio* energies is seen.

B. Properties of the CO_2 – H_2O PES

Stationary points of the full PES were obtained by applying Newton's method. The global minimum and four saddle point geometries are shown in Fig. 4. We confirm the earlier finding of Makarewicz³³ that there is only one minimum configuration denoted here as “a.” In “a,” the bond length for water in the complex is the same as the isolated monomer, the bond angle increases slightly to 105.1° . In addition, CO_2 is no longer linear in global minimum; instead, the $\text{O}=\text{C}=\text{O}$ angle is 178.1° with $\text{C}=\text{O}$ bond pointing away from H_2O . Note that R_{CO} for “a” is 2.7552 Å at the CCSD(T)/CBS(CP,AV5Z/AV6Z) level of theory;³⁴ the PES value of 2.753 Å is in good agreement with this benchmark value.

The PES stationary configuration energies are given in Table I, along with the present CCSD(T)-F12b/aVTZ *ab initio* calculations. The electronic dissociation energy D_e of “a” from the MP2/aVTZ PES³³ is 1004 cm^{-1} , and the benchmark CCSD(T)/CBS(CP,AV5Z/AV6Z) value

TABLE I. Energies of stationary points from indicated sources. Note that for the energy of configuration “a,” the global minimum is referenced to the isolated monomers and has the value of $-D_e$, while all energies of other configurations are referenced to “a.”

	a	b	c	d	e
CCSD(T)-F12b/aVTZ	−1054	297	332	619	877
PES	−1053	302	328	616	876
MP2/aVTZ PES ³³	−1004	291	308	635	828

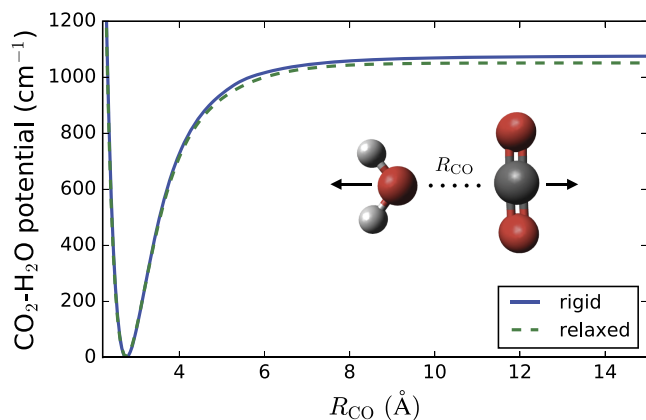


FIG. 5. Relaxed cut and rigid cut comparison for the dissociation path. Relaxed cut converges to 1053 cm^{-1} , while rigid cut converges to 1077 cm^{-1} . Both cuts have the same global minimum.

is 1024.8 cm^{-1} .³⁴ Thus, the PES value is in error by 28 cm^{-1} . The PES saddle point energies “b” to “e” are in excellent agreement with directly calculated ones, and the corresponding values from the rigid-monomer MP2/aVTZ-based PES³³ are in reasonable agreement with the present energies.

The fixed global minimum-configuration and relaxed monomer dissociation and internal rotation cuts from the PES are shown in Figs. 5 and 6, respectively. As expected, the fixed-monomer dissociation energy is larger than the relaxed one although the difference is small. There is a larger difference between the fixed and relaxed internal rotation barriers. These barrier heights are in reasonable agreement with empirically determined ones $305.36 \pm 0.5\text{ cm}^{-1}$ (fixed) and $285.66 \pm 1.4\text{ cm}^{-1}$ (relaxed).⁶² It is noteworthy that in the same research, the R_{CO} distance was reported as 2.83 Å which is 0.08 Å longer than the benchmark distance. Since there is relatively large discrepancy in the bond length between their and our research, it should not surprise us if there is also some discrepancy in internal rotation barriers.

The dissociation energy D_0 was obtained from D_e and the following DMC ZPEs: $2534.8 \pm 1.1\text{ cm}^{-1}$ for CO_2 , 4634.2

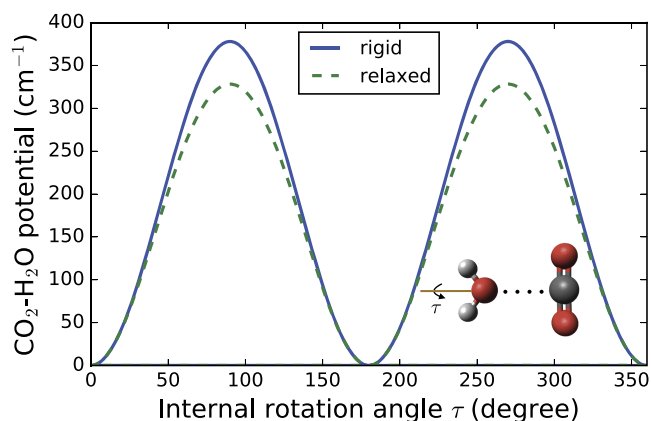


FIG. 6. Potentials for relaxed and rigid cuts versus the torsion angle τ , i.e., the angle between the vectors normal to the CO_2 and H_2O planes. The peak values are 328.0 and 378.1 cm^{-1} , respectively, for the relaxed and rigid cuts. Both cuts have the minimum potential of 0 when H_2O and CO_2 are coplanar and have the maximum potential when the H_2O plane is perpendicular to the CO_2 plane.

TABLE II. Harmonic frequencies (cm^{-1}) of $\text{CO}_2\text{--H}_2\text{O}$ calculated directly using CCSD(T)-F12b/aVTZ theory and labeled “*ab initio*” and from the PES.

Mode symbol	<i>Ab initio</i>	PES
$\nu_{\text{int}}(1)$	46	32
$\nu_{\text{int}}(2)$	75	83
$\nu_{\text{int}}(3)$	118	117
$\nu_{\text{int}}(4)$	174	159
$\nu_{\text{int}}(5)$	178	173
$\nu_{b1}(\text{CO}_2)$	664	664
$\nu_{b2}(\text{CO}_2)$	678	678
$\nu_{\text{sym}}(\text{CO}_2)$	1353	1355
$\nu_b(\text{H}_2\text{O})$	1646	1645
$\nu_{\text{asym}}(\text{CO}_2)$	2394	2400
$\nu_{\text{sym}}(\text{H}_2\text{O})$	3835	3830
$\nu_{\text{asym}}(\text{H}_2\text{O})$	3945	3952

$\pm 1.2\text{ cm}^{-1}$ for H_2O , and $7435.8 \pm 1.9\text{ cm}^{-1}$ for $\text{CO}_2\text{--H}_2\text{O}$. Thus, D_0 equals 787 cm^{-1} using a D_e of 1053 cm^{-1} and 758 cm^{-1} using the benchmark D_e . Unfortunately, no experimental value has been reported.

Harmonic frequencies of the global minimum obtained from both *ab initio* CCSD(T)-F12b/aVTZ calculations and the PES are presented in Table II. Before comparing these in detail, it should be kept in mind that the full PES is not a fit to CCSD(T)-F12b/aVTZ, but a composite consisting of a fit of 2-body CCSD(T)-F12b/aVTZ energies plus previous, spectroscopically accurate monomer potentials. With this in mind, we note some differences for some of the low frequency intermolecular modes. A likely explanation for this difference is that the minimum geometries are slightly different for the PES and *ab initio* calculations, and our experience is that small intermolecular low-frequency modes are more sensitive to small geometric changes than intermolecular modes. For the intramolecular modes, the PES results are within a few wavenumbers of the *ab initio* results. Also, the PES results are in very good agreement with previous CCSD(T)-F12b/aVTZ results.³⁴ Note that mode (3) is essentially the R_{CO} -stretch. The harmonic frequencies of the isolated monomers are given in parentheses next to the corresponding frequency in the dimer to indicate how these change in the dimer. For CO_2 , the degeneracy of the bend (at 673 cm^{-1}) is lifted in the dimer and, not surprisingly, the two frequencies, i.e., $\nu_{b1}(\text{CO}_2)$ and $\nu_{b2}(\text{CO}_2)$, straddle the isolated CO_2 one. With the notable exception of this bending mode, the monomer harmonic frequencies are only slightly changed upon complexation.

VSCF/VCI fundamental frequencies, calculated from MULTIMODE, are compared with experimental and other theoretical results in Table III. Note that there are two entries for $\nu_{\text{sym}}(\text{H}_2\text{O})$ with a small, i.e., 3 cm^{-1} , difference in energies. An examination of the VCI expansion coefficients reveals some mixing with a combination state consisting of one quanta of excitation each of the following: the lower-energy CO_2 bend $\nu_{b1}(\text{CO}_2)$, the symmetric CO_2 stretch, and the H_2O bend. This state is close in energy to the $\nu_{\text{sym}}(\text{H}_2\text{O})$ state and so mixing is not a surprise. (In first order, this mixing requires a 4-mode coupling matrix element.) Overall, the present results are in very good agreement with the experiment, especially the ones using

TABLE III. Comparison of fundamental frequencies of $\text{CO}_2\text{--H}_2\text{O}$ calculated from MULTIMODE, previous 2-mode anharmonic calculations of Ref. 14 and IR spectra from two matrix isolation experiments.

Mode	MULTIMODE	MP2/aVTZ ¹⁴	Ne matrix ⁶³	N_2 matrix ⁶⁴
$\nu_{b1}(\text{CO}_2)$	660	647	658.7	656.4
$\nu_{b2}(\text{CO}_2)$	676	659	672.2	665.3
$\nu_{\text{sym}}(\text{CO}_2)$	1285	1311	Not IR active	Not IR active
$\nu_b(\text{H}_2\text{O})$	1583	1487	1595.2	1597.0
$\nu_{\text{asym}}(\text{CO}_2)$	2348	2366	2348.2, 2348.8	2351.1
$\nu_{\text{sym}}(\text{H}_2\text{O})$	(3661, 3664)	3550	3664.8	3634.7
$\nu_{\text{asym}}(\text{H}_2\text{O})$	3759	3634	3760.3, 3754.5	3727.3

the Ne matrix. Also, they are substantially closer to the experiment than the previous calculations. This is probably mostly due to the higher level of *ab initio* theory used in the present calculations. However, the previous vibrational calculations were done using second-order perturbation theory corrections to VSCF energies and restricted to pairwise coupling of modes and so they are not as rigorous as the present ones. Finally, note that bending modes of CO_2 are strongly split, and there is virtually no Fermi resonance affecting the CO_2 symmetric stretch.

C. Application to $\text{CO}_2@(\text{H}_2\text{O})_{20}$

The fully optimized $\text{CO}_2@(\text{H}_2\text{O})_{20}$ geometry is shown in Fig. 7. The CO_2 bond angle is 179.8° , and both bond lengths are 1.160 Å. These values are very close to isolated CO_2 at equilibrium. The distances between C and the oxygen atoms in the surrounding water molecules are in the range 3.5–4.0 Å, substantially larger than this distance at the dimer minimum. So, clearly the equilibrium structure in the cage is a trade-off between attractive and repulsive interactions with the water molecules. Also, the orientation of the CO_2 bond axis is essentially perpendicular to and centered with respect to a pentagonal face.

From this minimum, we obtained an estimate of the electronic dissociation energy of 3799 cm^{-1} for CO_2 by using the same geometry for $(\text{H}_2\text{O})_{20}$ as for the global minimum and

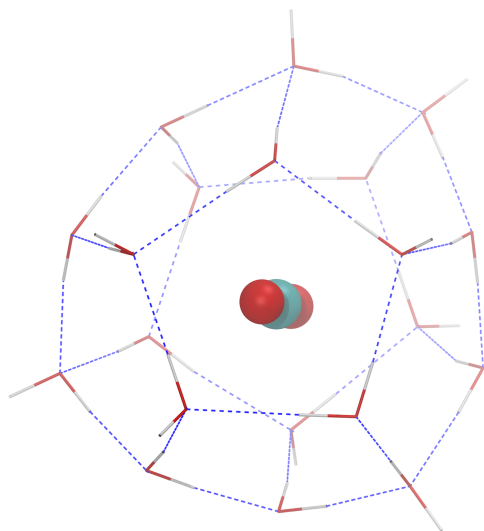


FIG. 7. Global minimum of flexible $\text{CO}_2@(\text{H}_2\text{O})_{20}$.

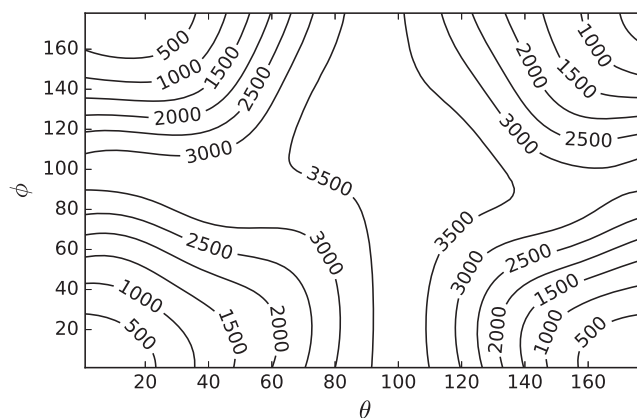


FIG. 8. Contour plot of the CO_2 rotational potential (in cm^{-1}) at the global minimum of $\text{CO}_2@(\text{H}_2\text{O})_{20}$ in the polar angles θ and ϕ (in degrees). The energy is referenced to global minimum, where θ and ϕ are zero.

for CO_2 removed. We also examined the least-motion path of CO_2 moving out of the cage through the center and perpendicular to the pentagonal face, with the water molecules held fixed. The barrier obtained is more than $20\,000\text{ cm}^{-1}$, and so clearly it would be necessary to open the pentagonal face for CO_2 to escape. In fact, this has been seen for the 5^{12} cage, in recent molecular dynamics calculations, using an empirical potential.⁵⁸

Figure 8 is a contour plot of the rotational potential for CO_2 at the global minimum position. For simplicity, CO_2 was set to be linear and the z -axis is the bond axis and the two angles θ and ϕ are the usual polar angles. The figure shows that the barrier is not small so the orientation of CO_2 is locked somewhat toward one of the pentagonal faces. The potential does have quasi-periodicity as there are twelve such faces (which, however, are not identical).

We performed a full normal-mode analysis of $\text{CO}_2@(\text{H}_2\text{O})_{20}$, and a smoothed density of states is plotted in Fig. 9. The smoothing was done by replacing each normal mode frequency (represented by a stick of constant height) by a Gaussian curve with a full-width half maximum of 10 cm^{-1} . Notice, under such resolution, the two CO_2 bending modes $\nu_{b1}(\text{CO}_2)$ and $\nu_{b2}(\text{CO}_2)$ are indistinguishable. Further, these modes are embedded in a similar spectral range of water cage

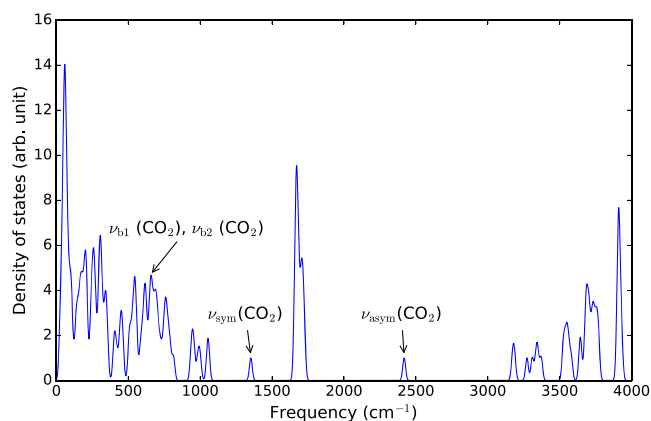


FIG. 9. Vibrational density of states of $\text{CO}_2@(\text{H}_2\text{O})_{20}$ calculated using the full-dimensional normal-mode analysis. Four high frequency intramolecular vibrations of CO_2 are labeled.

TABLE IV. 4-mode, 9-mode VSCF/VCI local-monomer vibrational energies and LMon and full harmonic frequencies (cm^{-1}) of the CO_2 monomer in $\text{CO}_2 @ (\text{H}_2\text{O})_{20}$.

Mode	4-mode	9-mode	LMon NMA	Full NMA
Frustr-trans	...	59	55	30
Frustr-trans	...	61	59	37
Frustr-rot	...	71	69	26
Frustr-rot	...	74	72	85
Frustr-trans	...	76	73	87
$\nu_{b1}(\text{CO}_2)$	654	656	660	655
$\nu_{b2}(\text{CO}_2)$	657	659	663	655
$\nu_{\text{sym}}(\text{CO}_2)$	(1292, 1359)	(1294, 1363)	1353	1352
$\nu_{\text{asym}}(\text{CO}_2)$	2356	2356	2418	2418

modes which suggests that some coupling of the CO_2 bending modes with cage modes is possible. However, an examination of the normal mode vectors shows this not to be the case. The symmetric and asymmetric CO_2 stretch modes are clearly isolated. So, this all suggests that a local monomer treatment of the CO_2 intramolecular vibrational modes should be reasonable. This is verified in Table IV, where the local-monomer harmonic frequencies of CO_2 are compared with corresponding results from the full normal-mode analysis. There are larger differences for the bend modes than the stretches, and this is understandable, based on the preceding discussion of the vibrational density states.

VSCF/VCI vibrational energies using just the four high-frequency intramolecular modes and all (9) local monomer modes are given in this table as well. As seen, there is very good agreement between these two sets of results, implying little coupling between 5 low-frequency modes and 4 high-frequency modes. Both sets of calculations find strong mixing between the symmetric stretch with the bend which results in a Fermi-diad, analogous to the isolated gas-phase Fermi-diads at 1286 and 1383 cm^{-1} . These Fermi-diads have been observed in a room temperature Raman spectrum of CO_2 in the larger $5^{12}6^2$ hydrate clathrate at roughly 1277 and 1382 cm^{-1} ,⁶⁵ values that are close to the gas-phase ones. The present calculations show an up-shift of the lower-energy diad and a down-shift of the higher energy one, relative to these experimental results, which unfortunately are not directly comparable to the present calculations. The calculated CO_2 asymmetric stretch at 2356 cm^{-1} is in good agreement with an experimental value of 2347 cm^{-1} , obtained in IR reflectance spectra of CO_2 in the smaller of two clathrate hydrates.⁶⁶

Finally, comparing VSCF/VCI calculations for $\text{CO}_2 @ (\text{H}_2\text{O})_{20}$ shown in Table IV with calculations for the $\text{CO}_2\text{—H}_2\text{O}$ system in Table III, one can find that two bending modes are much less separated in $\text{CO}_2 @ (\text{H}_2\text{O})_{20}$. The energy separation is about 14–16 cm^{-1} for $\text{CO}_2\text{—H}_2\text{O}$ while it is only about 1–3 cm^{-1} for $\text{CO}_2 @ (\text{H}_2\text{O})_{20}$. Also, the symmetric stretch is not strongly perturbed by the bend in the dimer, where it is in the hydrate clathrate. The reason for these significant differences is likely because the $\text{CO}_2\text{—H}_2\text{O}$ interaction is much stronger in the dimer than in the hydrate clathrate (recall the differences in the distance) in the two so the perturbation is smaller. In addition, the anisotropy is much less in $\text{CO}_2 @ (\text{H}_2\text{O})_{20}$ since CO_2 is perturbed roughly uniformly from

all directions. Five non-zero low-frequency modes show that CO_2 is unable to rotate or translate freely in the water cage. It is noteworthy that in full-scale normal-mode analysis, these five frustrated rotational and translational modes plus two bending modes are all coupled significantly with water cage movements so that the local-monomer model cannot recover them accurately.

IV. SUMMARY AND CONCLUSIONS

In this article, we presented an accurate, full-dimensional, 2-body potential energy surface for $\text{CO}_2\text{—H}_2\text{O}$. This potential energy surface is a hybrid of two fits, denoted fit-SR and fit-LR linked by a switching function. fit-SR is a permutationally invariant fit to 47 593 *ab initio* energies of CCSD(T)-F12b/aVTZ level theory. fit-LR is also a permutationally invariant fit to 6244 CCSD(T)-F12b/aVTZ electronic energies, that rigorously goes to zero in the long range and which has an RMS fitting error of 0.4 cm^{-1} . This 2-body potential energy surface is combined with previous spectroscopically accurate potentials for the monomers to produce the final potential energy surface. The utility of this 2-body potential was demonstrated by using it, together with a previous water potential, to develop a potential for the small CO_2 hydrate clathrate $\text{CO}_2 @ (\text{H}_2\text{O})_{20}$. Numerous properties of these potentials, including stationary points, harmonic frequencies, etc., were reported. The rigorous dissociation energy, D_0 , of the dimer was reported as 787 cm^{-1} using the 2-body PES D_e of 1053 cm^{-1} and 758 cm^{-1} using the benchmark D_e . The latter value is then the current benchmark number for this important dimer. VSCF/VCI coupled-mode calculations of vibrational fundamentals were reported for the seven intramolecular modes of the dimer and the local-normal modes of CO_2 in the hydrate clathrate, using a 4-mode representation of the potential for both sets of calculations. These were compared, with no scaling or empirical adjustments, with available experimental results, and agreement within a few wavenumbers was seen.

In summary, the new 2-body potential is a major step forward in providing both a highly accurate and transferable potential that should see many future applications.

SUPPLEMENTARY MATERIAL

See [supplementary material](#) for details on downloading the software to evaluate the 2-body potential energy surface.

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